

NARWC Database Project

Northern Atlantic Right Whale Consortium aerial survey database management system.

Status: Phase 1 in progress — historical migration pipeline functional; lookup table gaps and per-survey corrections blocking full migration run.

Quick Start

```
% 1. Initialize project
startup

% 2. Configure database (one-time)
% Copy config/local/db_config_local.m.template → config/local/db_config_local.m
% and fill in your credentials.

% 3. Test connection
scripts/setup/test_connection

% 4. Use database
conn = narwc.db.Connection.create();
data = conn.fetch('SELECT TOP 10 * FROM Master');
conn.close();
```

Requirements

- MATLAB R2020b or later
- Database Toolbox
- SQL Server, MySQL, or PostgreSQL

Project Structure

```
NARWC-DB/
├─ startup.m                # Add paths, check toolboxes, create data dirs –
run first
├─ config/
│   ├─ load_config.m        # Layered config: defaults < local < batch
│   ├─ defaults/           # Baseline values (db, validation, pipeline)
│   └─ local/               # Gitignored –
db_config_local.m with credentials
│   └─ batches/             # Per-
run overrides: migration.m (permissive), routine.m (strict)
│       └─ overrides/        # Per-batch warning-
acknowledgement CSVs (migration_overrides.csv, routine_overrides.csv)
```

```

├── src/
│   ├── +migration/
│   │   └── apply_known_fixes.m          # Category C pre-
validation corrections
│   ├── +narwc/
│   │   ├── +ingestion/                # BatchUploader, SurveyExtractor, SurveyFileWriter,
│   │   │                                       # convert_contributor_batch, run_batch_upload
│   │   ├── +db/                      # Connection, FieldDefinitions
│   │   ├── +io/+parsers/             # StandardFormat, NEAQFormat, TemplateFormat, ParserFact
│   │   ├── +validation/             # SurveyValidator, ErrorCollector, +rules/
│   │   ├── +processing/             # SurveyProcessor, ChangeTracker, +steps/
│   │   └── +reports/                # ValidationReport, ProcessingReport, SummaryStatistics
├── scripts/
│   ├── migration/                    # Steps 0-
3 + run_full_migration.m (one-time legacy migration)
│   ├── ingestion/                   # convert/upload_contributor_batch (routine per-
contributor ingestion)
│   ├── sql/                         # Schema, curation, verification, teardown scripts
│   └── setup/                       # test_connection, pull/push_lookup_tables
├── tests/                          # test_runner.m, fixtures/, unit/
├── lib/+logging/                   # Logging toolbox
├── data/                          # Runtime data (mostly gitignored)
│   └── tables/                     # Lookup table CSVs (committed)
├── docs/                          # Developer guide, config reference, override guide
└── ref/                          # NARWC Users Guide PDF and reference material

```

Common Commands

```

startup                                % Add all paths, check toolboxes, create data di
scripts/setup/test_connection          % Verify database connection

% Validation
validator = narwc.validation.SurveyValidator();
[is_valid, results] = validator.validate(data);

% Migration (run in order)
config = load_config('migration');
step1_extract_surveys(csv_file)
step2_upload_surveys('Config', config)
step3_validate_migration()

% Tests
test_runner()                          % Run all tests
test_runner('unit', 'Verbose', true) % Unit tests only, verbose

```

Development Status

- ☒ Phase 0: Project infrastructure, database connectivity
- ☒ Phase 0: Validation framework (9 rule modules, override system)
- ☒ Phase 0: Migration pipeline (extract → validate → upload, transaction-safe)
- ☐ Phase 1: Historical migration — pipeline ready; blocked on lookup table gaps and per-survey corrections
- ☐ Phase 2: New-survey ingestion (NEAQFormat parser, curator GUI)

See `PROJECT_STATUS.md` for the full active work list, open questions, and per-survey data issues.

Known Data Issues

The legacy CSV contains a number of data quality issues that require curator decisions before or during migration. These are tracked in `PROJECT_STATUS.md` §8.5 and fall into three categories:

- Lookup table gaps — missing SPECCODE, BEHAV, ANHEAD, PLATFORM, BLOCK, and GLARE codes; tracked in `PROJECT_STATUS.md` §8.4
- Category C bulk corrections — sentinel values (99), PHOTOS=0, STRIP>16 handled automatically by `apply_known_fixes.m`
- Per-survey manual corrections — specific event/field errors in individual surveys requiring curator or Bob sign-off; tracked in `PROJECT_STATUS.md` §8.5

Documentation

- Configuration: `docs/configuration_reference.md`
- Validation rules and overrides: `docs/warning_overrides.md`
- Testing: `docs/testing_guide.md`
- Database schema: `docs/database_schema.md`
- Help: `help narwc`, `help narwc.db.Connection`

Contact

russ.shomberg@marineacoustics.com

Version: 0.2.0 | Last Updated: 2026-06